

Animal Class Test Strategy

1. Create an Animal Object with the default constructor of Animal class.
2. Create an Animal Object with the non-default constructor of Animal class.
Test cases for non default constructor
 - a. valid fields
 - b. invalid fields
3. Test cases for one get methods: getIdentity()
 - a. valid field values
 - b. invalid field values
4. Test cases for one set methods: setIdentity()
 - a. valid field values
 - b. invalid field values
5. Test case for toString() method to check display of Animal data.

Test 1:

Creation of Animal object with default constructor i.e. `Animal animal = new Animal();`

Test data:

identity: 0
alive: true
spotted: false
species: default species object
 name = "unknown"
 type = "unknown"
 diet = "unknown"
 predator = false
 aggressionLevel = "unknown"

Expected results:

Animal Status:

Identity: 0
Alive: Yes
Spotted: No

Species:

Name: unknown
Type: unknown

Diet: unknown
Predator: No
Aggression Level: unknown

Actual Result:

```
Test Case 1: Default Constructor    -----Animal animal1 = new Animal()

Animal Status:
    Identity: 0
    Alive: Yes
    Spotted: No

Species:
    Name: unknown
    Type: unknown
    Diet: unknown
    Predator: No
    Aggression Level: unknown
```

Test 2a:

Creation of Animal object with non-default constructor with the valid field values.

Test data:

identity = 111
alive = true
spotted = false
species: valid Species object
 name = "Thylacoleo"
 type = "marsupial"
 diet = "carnivore"
 predator = true
 aggressionLevel = extreme

Expected result:

Animal Status:
 Identity: 111
 Alive: Yes
 Spotted: No
Species:
 Name: Thylacoleo
 Type: marsupial
 Diet: carnivore
 Predator: Yes

Aggression Level: extreme

Actual results:

```
Test Case 2a: Parameterized Constructor for valid fields  -----Animal animal2 = new Animal(111, species, true, false)

Animal Status:
  Identity: 111
  Alive: Yes
  Spotted: No

Species:
  Name: Thylacoleo
  Type: marsupial
  Diet: carnivore
  Predator: Yes
  Aggression Level: extreme
```

Test 2b:

Creation of Animal object with non-default constructor with the invalid field values.

Test data:

identity = 99
alive = true
spotted = false
species: valid Species object
 name = "Thylacoleo"
 type = "marsupial"
 diet = "carnivore"
 predator = true
 aggressionLevel = extreme

Expected results:

Error: Enter valid ID between range 100 to 999

Animal Status:

Identity: 0
Alive: Yes
Spotted: No

Species:

Name: Thylacoleo
Type: marsupial
Diet: carnivore
Predator: Yes
Aggression Level: extreme

Actual results:

```
Test Case 2b: Parameterized Constructor for invalid fields -----Animal animal3 = new Animal(99, species, true, false)
```

```
Error: Enter a valid 3 digit ID i.e 001 to 999 INT
```

```
Animal Status:
```

```
Identity: 0
```

```
Alive: Yes
```

```
Spotted: No
```

```
Species:
```

```
Name: Thylacoleo
```

```
Type: marsupial
```

```
Diet: carnivore
```

```
Predator: Yes
```

```
Aggression Level: extreme
```

Test 3a:

Testing getIdentity() method with valid identity value using animal object.

Test data:

Testing via already created animal object created i.e animal2

Expected result: 111

Actual result:

```
Test Case 3a: getIdentity() for valid value: -----Using animal2 object  
111
```

Test 3b :

Testing getIdentity() method with invalid identity value using animal object.

Test data:

Testing via already created animal object created i.e animal1

Expected result O/P:

0

Actual result:

```
Test Case 3b: getIdentity() for invalid value: -----Using animal1 object  
0
```

Test 4a:

Testing setIdentity() method with valid identity value.

Test data:

Animal object updated using method call i.e animal.setIdentity(100)

Expected: identity = 100

Actual:

```
Test Case 4a: setIdentity(100) for valid value:  -----animal1.setIdentity(100)
100
```

Test 4b:

Testing setIdentity() method with invalid identity value.

Test data:

Method called: animal.setIdentity(-100) for invalid value error message displayed

Expected result:

Error message: "Error: Enter valid ID between range 100 to 999"

0

Actual result:

```
Test Case 4b: setIdentity(-100) for invalid value:  -----animal1.setIdentity(-100)

Error: Enter a valid 3 digit ID i.e 001 to 999 INT
0
```

Test 5:

Testing toString() method to check display of Animal and Species data.

Test data:

identity = 111

alive = true

spotted = false

species: valid Species object

name = "Thylacoleo"

type = "marsupial"

diet = "carnivore"

predator = true

aggressionLevel = extreme

Expected result:

Animal Status:

Identity: 111

Alive: Yes

Spotted: No

Species:

Name: Thylacoleo

Type: marsupial

Diet: carnivore

Predator: Yes

Aggression Level: extreme

Actual result:

```
Test Case 5: toString():
```

```
Animal Status:
```

```
    Identity: 111
```

```
    Alive: Yes
```

```
    Spotted: No
```

```
Species:
```

```
    Name: Thylacoleo
```

```
    Type: marsupial
```

```
    Diet: carnivore
```

```
    Predator: Yes
```

```
    Aggression Level: extreme
```